

Interfacing and Use Cases for a Five-Finger Dexterous Hand Attached to a UR5e: A Design and Evaluation-Plan Study

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Abstract—Five-finger dexterous hands extend a collaborative arm beyond the fixed grasp set of a parallel-jaw gripper, but attaching such a hand to a specific arm raises concrete mechanical, electrical, and software-integration questions. This paper studies the integration of an Inspire Robots RH56F1-E2R-T1 dexterous hand—a right-hand, tactile-equipped model with an EtherCAT and RS-485 control interface—onto a Universal Robots UR5e collaborative arm. Using only vendor-published specifications, the paper characterises both platforms and derives an integration design. The central electrical finding is a supply mismatch: the hand draws up to 5 A at 24 V, whereas the UR5e tool connector supplies at most 1.5 A (dual-pin). Two power concepts are compared—a flange-integrated configuration that respects the tool-connector budget through register-level current limiting, and an externally powered configuration that preserves the hand’s full force envelope while routing RS-485 through the robot’s Tool Communication Interface (TCI). The communication path is realised with the Universal Robots tool-communication forwarding URCap and ROS 2 driver, which relay the RS-485 channel to a virtual serial device on the control PC. The paper then surveys five use-case families enabled by the combined system, specifies a three-stage evaluation protocol, and analyses the principal integration risks. No physical measurements have been collected at the time of writing: the hardware has been ordered but not yet integrated, so all experimental results are explicitly pending and the contribution is an integration design and a pre-registered evaluation plan.

Index Terms—dexterous robotic hand, five-finger hand, UR5e, RS-485, tool communication interface, tactile sensing, grasping, ROS 2, human-robot interaction

I. INTRODUCTION

Most industrial and collaborative robot arms are equipped with two-finger parallel grippers or vacuum tools. These end-effectors are robust and simple to control, but they restrict the arm to a small family of grasps: an object must present two roughly parallel faces, the grasp must be planned so that the closing axis of the gripper aligns with those faces, and once grasped, an object cannot be re-oriented within the gripper. A human hand supports a far larger set of grasp types, and everyday manipulation—operating tools, handling containers, passing objects between people—draws on that variety [6]. Multi-fingered hands combined with contact sensing have

therefore long been identified as a route toward more general manipulation [11].

Commercial dexterous hands have recently become easier to integrate. The RH56 family from Beijing Inspire-Robots is one example: the hands are anthropomorphic, five-fingered, electrically actuated, and controlled through serial interfaces rather than proprietary controllers [1]. Hands from this manufacturer have been used as end-effectors in immersive teleoperation research [7]. In parallel, the Universal Robots e-Series has become a common research arm with open interfaces, a maintained ROS 2 driver, and—relevant here—an RS-485 serial channel at the tool flange [4], [5].

This work concerns a concrete integration task in the autonomous-systems laboratory at Ostfalia: attaching an Inspire Robots RH56F1-E2R-T1 dexterous hand (right hand, tactile sensors, RS-485 variant) to a UR5e arm. Two questions are addressed. First, how should the hand be connected—mechanically, electrically, and in software—so that arm and hand can be driven from one control stack? Second, which application classes does the combined arm–hand system enable that a parallel gripper does not, and which are feasible in a student-project setting?

The scope of the paper is deliberately limited to what can be established without fabricated data. At the time of writing the hardware has been ordered but not yet received or integrated; consequently the paper does not report any measured results. Its contributions are: (i) a specification-level characterisation of the two platforms drawn from vendor documentation; (ii) an integration design covering mechanics, power, communication, and software, including identification of a supply-current mismatch and two concrete ways to resolve it; (iii) a survey of use cases mapped onto the verified capabilities of the hardware; and (iv) a pre-registered three-stage evaluation protocol whose execution is future work. Section II reviews related work; Section III characterises the hardware; Section IV develops the integration design; Section V surveys use cases; Section VI specifies the planned evaluation; Section VII analyses integration risks; and Sections VIII–XI discuss the design, its limitations, future work, and conclusions.

II. RELATED WORK

A. Dexterous Manipulation and Grasp Taxonomies

Multi-fingered hands span a wide design space, from highly articulated research hands to underactuated designs that trade degrees of freedom for robustness and cost. Billard and Kragic review the field and argue that manipulation—and in particular the coupling of dexterous kinematics with tactile feedback—remains a central challenge: robots handle known rigid objects in structured settings but degrade as object compliance and contact conditions vary [11]. For assessing a hand’s grasp repertoire, Feix et al. provide a widely used reference: they consolidate prior taxonomies into a single scheme of 33 human grasp types, organised by properties such as power versus precision and thumb role [6]. This taxonomy is adopted later as a checklist for the RH56F1’s achievable grasps and as vocabulary for the evaluation protocol. A limitation of taxonomy-based analysis is that it describes human grasps; an underactuated six-DOF hand realises only a subset, so the mapping is used descriptively rather than as a claim of full coverage.

B. Tactile Sensing for Grasping

Dahiya et al. survey tactile sensing for robots and argue that stable grasping and safe interaction need distributed contact information, which joint-level torque cannot supply: actuator torque neither localises a contact nor detects incipient slip [8]. Their survey predates the current generation of hands with integrated sensing, and it treats tactile skin largely as an add-on subsystem. The RH56F1 instead integrates tactile pads that report over the hand’s own bus, so force-threshold grasping—closing until a target contact force is sensed—is available without external sensor electronics [1]. The present work relies on this integrated sensing but does not contribute new tactile-processing methods.

C. Teleoperation and Learning from Demonstration

Dexterous hands on serial arms are the standard substrate for vision-based teleoperation. DexPilot demonstrated markerless, glove-free teleoperation of a multi-finger hand from depth-camera hand tracking [9]; AnyTeleop generalised this approach into a vendor-neutral system supporting multiple arms and hands [10]; and Open-TeleVision combined VR stereo streaming with hand retargeting, using Inspire hands as its end-effectors [7]. These systems retarget tracked human hand poses onto the robot hand and are today primarily used to collect demonstrations for imitation learning. A common limitation is that they are demonstrated on the authors’ own hardware stacks, and the mapping from a tracked human hand to a particular robot hand must be re-derived per device. Their relevance here is architectural: arm and hand are controlled by separate drivers at different rates, and a thin coordination layer on one PC suffices—the same structure adopted in Section IV. This paper does not implement teleoperation; it identifies teleoperation as a supported use case given the hand’s register-addressable joints.

TABLE I
POSITIONING OF THIS WORK AGAINST THE REVIEWED LITERATURE

Work	Focus	Hand / plat-form	Tactile in loop
Feix et al. [6]	Grasp taxonomy	Human hands	—
Dahiya et al. [8]	Tactile survey	—	Survey scope
DexPilot [9]	Vision teleop.	Multi-finger, arm	No (vision)
AnyTeleop [10]	Teleop. framework	Multiple hands/arms	No (vision)
Open-TeleVision [7]	Immersive teleop.	Inspire, humanoid	No (vision)
This paper	Integration design	RH56F1 on UR5e	Planned (Stage 3)

D. End-Effector Integration on Universal Robots Arms

The e-Series exposes an RS-485 Tool Communication Interface at the tool flange, so that a serial device mounted at the tool centre point can be reached without running a separate data cable along the arm [4], [5]. Universal Robots publishes an open-source URCap that relays this channel to a remote PC, and the ROS 2 driver integrates the mechanism through launch parameters [4], [5]; the resulting path is described in Section IV-C. Practitioner documentation notes concrete pitfalls, including a reported conflict when the RS-485 forwarding URCap and certain gripper URCaps run in parallel [4]. These sources establish the integration pathway but describe it generically; they do not address a specific hand such as the RH56F1, nor its power requirements. No published integration of the RH56F1-E2R-T1 with the UR5e was identified during the literature review; addressing this gap at the design level is the aim of this paper.

Table I positions the present work against the sources reviewed above, along the axes that matter for this project: the hand and platform used, and whether tactile information forms part of the control loop. The three teleoperation systems are all vision-driven and are demonstrated on their authors’ own platforms; none documents attachment to a UR e-Series tool interface, and none closes a control loop on tactile data.

III. HARDWARE PLATFORMS

A. Inspire Robots RH56F1-E2R-T1 Dexterous Hand

The RH56F1 is a five-finger anthropomorphic hand with six actuators driving twelve mechanical joints; it is therefore underactuated, with each of the four fingers driven in flexion by a single actuator and the thumb driven by two (flexion and lateral rotation) [2]. According to the manufacturer, the RH56F1 series has an integrated metal skeletal structure and a passive (structural) load capacity of the order of 30 kg [2]. The series is offered in four build variants that differ in control interface and sensing, summarised in Table II; for this project

TABLE II
RH56F1 SERIES BUILD VARIANTS (RIGHT-HAND MODELS; LEFT-HAND EQUIVALENTS ALSO EXIST)

Variant	Control interface	Tactile sensing
E2R-T1 (selected)	EtherCAT + RS-485	8 sensors
E2R	EtherCAT + RS-485	None
E4R-T1	EtherCAT + CAN FD	8 sensors
E4R	EtherCAT + CAN FD	None

TABLE III
PUBLISHED PARAMETERS OF THE RH56F1-E2R-T1 [1], [2]

Parameter	Value
Control interface	EtherCAT + RS-485
Degrees of freedom / joints	6 / 12
Weight	630 $\pm$ 10 g
Operating voltage	24–48 V
Quiescent current (RS-485, 24 V)	194 $\pm$ 10 mA
Peak current (24 V)	5 A
Repeatability	$\pm$ 0.2 mm
Thumb fingertip strength	$\geq$ 15 N
Finger fingertip strength	$\geq$ 10 N
Force resolution	0.1 N
Thumb lateral rotation range	60–180°
Thumb lateral rotation speed	$>$ 155°/s
Thumb flexion speed	$>$ 37°/s
Four-finger flexion speed	$>$ 107°/s
Tactile sensors	8
Tactile range / accuracy	30 N (FS) / 5% FS
Hand closing time	$\leq$ 0.8 s

the choice of variant is constrained by the target arm rather than free.

Two requirements fix the variant. First, attachment through the UR5e tool flange uses the RS-485 Tool Communication Interface (Section III-B); of the four variants only the E2 pair exposes RS-485, whereas the E4 pair uses CAN FD, which the standard tool connector does not carry. Second, the force-controlled grasping use case (Section V-B) requires the tactile-equipped “-T1” build. Together with the ordered right-handedness, these constraints select the RH56F1-E2R-T1 uniquely. Its published parameters are listed in Table III [1], [2]. The hand family itself was procured for the laboratory rather than selected by this study; the scope here is to analyse its integration, not to re-open the comparison against more articulated research hands, which differ in cost, control complexity, and support burden. Hardware selection is therefore outside the scope of this work.

Two aspects of Table III drive the integration. First, the electrical envelope: the hand accepts 24–48 V, draws about 0.19 A at 24 V when quiescent, and specifies a peak of 5 A; the implications of this peak for the UR5e tool connector are analysed in Section IV-B. Second, the sensing relevant to this project is eight tactile sensors reporting up to 30 N (full scale) with 5% accuracy, together with a 0.1 N force resolution; a per-

TABLE IV
PUBLISHED PARAMETERS OF THE UR5e [3]

Parameter	Value
Payload / reach	5 kg / 850 mm
Pose repeatability	$\pm$ 0.03 mm (ISO 9283)
Degrees of freedom	6
Tool flange	EN ISO 9409-1-50-4-M6
Tool connector	M8, 8-pin
Tool I/O supply voltage	12 / 24 V
Tool I/O supply current	1.5 A (dual pin), 1 A (single pin)
Tool serial channel	RS-485 via TCI
Arm weight (incl. cable)	20.6 kg

DOF force-threshold register lets closing halt once the sensed force reaches the set value, which is the basis for grasping objects of differing hardness [1]. The hand is described by the manufacturer as power-off self-locking—that is, designed to retain a grasp without continuous motor power [1].

The control interface warrants comment. The E2 variant exposes both EtherCAT [14] and RS-485 [2]. For attachment through the UR5e tool flange, the RS-485 channel is the usable one, because the flange’s Tool Communication Interface is itself RS-485 (Section III-B); EtherCAT would require cabling that the standard tool connector does not provide. Inspire’s RH56-series hands are commanded through a register-based serial protocol documented for the series in the user manual [1]; the exact register map for the RH56F1 is defined in its own manual, which the integration will follow rather than assume.

B. Universal Robots UR5e

The UR5e is a six-axis collaborative arm with a 5 kg payload, 850 mm reach, and $\pm$ 0.03 mm pose repeatability per ISO 9283 [3]. As a collaborative robot it provides certified force- and speed-limiting functions that permit fenceless operation, which is relevant to the human–robot-interaction use cases in Section V. The parameters that matter for tool integration appear in Table IV [3].

The tool connector is central to the integration. It provides 12 V or 24 V with a current limit of 1.5 A when two supply pins are paired and 1 A in single-pin mode [3]. Its serial capability is the Tool Communication Interface, an RS-485 bus presented on the tool flange for attaching an RS-485 device at the tool centre point without a separate cable [4], [5]. The mechanical interface is an EN ISO 9409-1-50-4-M6 flange with an M8 8-pin connector [3]. The e-Series therefore provides the physical layer the RH56F1-E2R-T1 expects on its RS-485 side; whether the electrical budget and the software path also fit is examined next.

IV. SYSTEM DESIGN AND INTERFACING

A. Mechanical Mounting and Payload

The hand’s wrist interface differs from the UR5e’s EN ISO 9409-1-50-4-M6 flange, so a passive adapter plate is expected

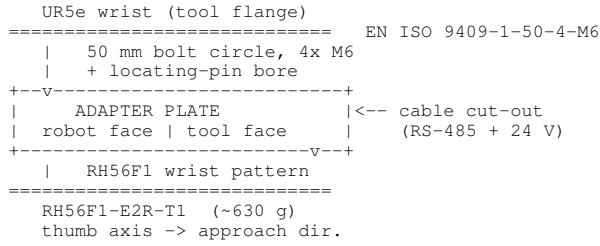


Fig. 1. Conceptual side view of the mechanical stack-up. The passive adapter plate mates the arm’s EN ISO 9409-1-50-4-M6 flange to the hand’s wrist interface; a lateral cut-out routes the hand’s power and RS-485 leads clear of the mating faces. Dimensions are not to scale, and the hand-side pattern is taken from the hand’s mechanical drawing.

to be required: on the robot side it reproduces the 50 mm four-hole M6 bolt circle with the locating-pin bore of EN ISO 9409-1, and on the hand side it matches the RH56F1 wrist interface. At about 630 g plus adapter and cabling, the assembled tool mass is far below the 5 kg payload of the UR5e [2], [3], leaving margin for a grasped object. Two robot settings must be maintained regardless of the concept chosen: the payload mass and centre of gravity, used by the e-Series dynamics and safety model, and the tool-centre-point offset, which should sit at a task-meaningful location such as the grasp volume between thumb and fingers. Because the ordered unit is a right hand, the thumb defines a preferred approach direction; fixing and documenting the mounting angle, and encoding it consistently in the TCP definition and the URDF model, prevents later mismatches between simulation and hardware.

Fig. 1 shows the intended stack-up in schematic side view. The plate is a two-sided part: its robot face carries the standardised flange pattern, its tool face carries the hand’s wrist pattern, and a lateral cut-out routes the RS-485 and power leads out of the joint so they are not pinched between the mating faces. Thickness is set by clearance for the connector and the cable bend radius rather than by strength, since the loads are small relative to the plate’s section.

B. Electrical Integration: Two Power Concepts

The primary integration constraint is electrical. The hand accepts 24 V, which the tool connector can supply, but its peak draw of 5 A exceeds the 1.5 A dual-pin limit of the tool supply [2], [3]. The gap is best seen in power terms: the tool connector can deliver at most $24 \text{ V} \times 1.5 \text{ A} = 36 \text{ W}$, whereas the hand’s specified peak is $24 \text{ V} \times 5 \text{ A} = 120 \text{ W}$, so the flange can supply only about 30% of the hand’s peak electrical power. The quiescent draw is roughly $24 \text{ V} \times 0.19 \text{ A} \approx 4.6 \text{ W}$. Assuming the quiescent electronics draw remains approximately constant during motion, about 31 W then remains for actuation inside the flange budget—sufficient for low-force or sequential finger motion, but not for simultaneous high-force actuation of all six DOFs. The actual draw is expected to depend on the number of simultaneously actuated joints and the operating load, and is quantified in Stage 1 of the evaluation (Section VI). Two concepts follow; Table V compares them.

TABLE V
COMPARISON OF THE TWO POWER CONCEPTS

Aspect	Concept A	Concept B
Power source	24 V tool supply	External $\geq 5 \text{ A}$ supply
Communication	RS-485 via TCI	RS-485 via TCI / USB
External cabling	None	One power cable
Peak grip force	Reduced (limited)	Full ($\geq 15 \text{ N}$ thumb)
Closing speed	Reduced	Full
Main risk	Tool over-current trip	Cable / shared ground
Suited for	Light grasping	Force experiments

Concept A — flange-integrated. The hand is powered from the 24 V tool supply and communicates over the TCI. No cable runs along the arm, which is mechanically clean and preserves the arm’s motion range. The 1.5 A budget must then be enforced by configuration: the RH56-series register interface exposes current-limit and power-on speed/force settings [1], which can be set so that worst-case simultaneous actuation stays within budget. The costs are reduced peak grip force and slower closing, and the risk that an unanticipated over-current trips the tool supply. Because the hand’s peak is roughly triple the available current, this concept is viable only for light, low-force grasping and must be validated by measurement before use.

Concept B — external supply, flange communication. The hand’s 24 V power comes from a dedicated supply rated at 5 A or more, routed along the arm with a strain-relieved cable, while RS-485 still uses the TCI (or, as a fallback, a USB-RS-485 converter at the PC). This preserves the full force and speed envelope at the cost of one external cable and a shared-ground requirement: the external supply’s 0 V must be bonded to the tool-connector ground so the RS-485 differential pair stays within its common-mode range.

For this project Concept B is proposed as the default, because grasp-force experiments are among the intended use cases and derating the hand would confound them. Concept A is retained as a demonstrator of the mechanically simplest possible integration and can be attempted once the register-level current limiting is understood and the actual current profile of the hand has been measured (Section VI). This choice is a design decision, not a measured result.

C. Communication Architecture

When the TCI is used, three layers connect a program on the control PC to the hand. First, the UR5e Tool I/O is switched from analog-input mode to communication-interface mode and set to the hand’s serial parameters in the robot’s installation. Second, the Universal Robots tool-communication forwarding URCap runs a relay on the robot controller that exposes the RS-485 device over TCP port 54321 [5]. Third, on the control PC a `socat` process (or the equivalent option in the ROS 2 driver) binds that TCP stream to a virtual serial device such as `/tmp/ttyUR`, which client software opens like a local RS-485 adapter [4], [5]. Fig. 2 shows the chain.

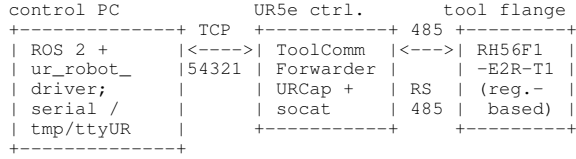


Fig. 2. Communication path for the flange-integrated configuration. The hand’s RS-485 control interface is relayed from the tool flange to a virtual serial device on the control PC over TCP port 54321, following the Universal Robots tool-communication forwarding mechanism [4], [5]. Serial parameters are set to match the hand’s manual.

On top of the virtual device, hand control reduces to register access over RS-485: a posture is commanded by writing the per-DOF target-angle registers, and feedback is obtained by reading the actual-angle and sensed-force registers [1]. A small driver wrapping the register addresses defined in the hand’s manual is sufficient, and named postures (open, power grasp, pinch, tripod, point) can be stored as six-value angle vectors. Specific register addresses and the serial baud rate are taken from the RH56F1 manual and are not reproduced here, to avoid asserting values that this study has not independently confirmed.

A first-order bandwidth check indicates that register polling is unlikely to be the bottleneck. At a representative RS-485 rate of 115200baud with 8N1 framing, each byte occupies about 87 μ s. A write of six 16-bit target registers, together with the address, function-code, and checksum fields, is on the order of 20–30 bytes depending on the register transaction, i.e. roughly 2–3 ms per transaction; a feedback read of comparable length is similar. The theoretical link budget therefore leaves ample margin for a 50 Hz command-and-feedback loop. This estimate is illustrative rather than a specification: the exact rate is configured per the hand’s manual, and the achievable throughput will depend on the relay latency and software scheduling, which are among the quantities to be measured in Stage 1 (Section VI).

D. Software Stack and Arm–Hand Coordination

The arm is driven by the official Universal Robots ROS2 driver [4], which builds on ROS2 [12], provides joint-trajectory and Cartesian control, and integrates with MoveIt 2 [13] for motion planning. The same driver manages tool-communication forwarding: launching it with `use_tool_communication` set true and the matching serial parameters creates the virtual serial device automatically [4]. The proposed stack runs on one Linux PC: the UR ROS2 driver for the arm, a lightweight hand node exposing services such as `set_posture` and `grasp_with_force` over the register layer, and an application layer sequencing arm motions and hand actions. For planning, the hand is added to the arm’s URDF as a fixed link with an approximate collision mesh; modelling individual finger joints is optional and needed only for finger-level collision checking. Fig. 3 shows the resulting stack.

Hard real-time coupling between arm and hand is not required for the target use cases: a grasp is structured as move-

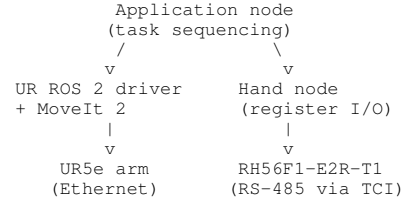


Fig. 3. Software stack on the single control PC. The application node sequences arm and hand actions; the arm is commanded through the Universal Robots ROS2 driver with MoveIt 2, and the hand through a thin node performing register reads and writes over the virtual serial device of Fig. 2.

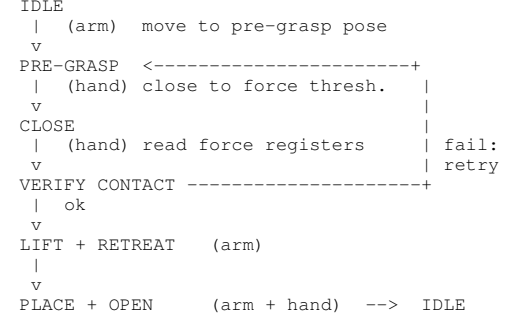


Fig. 4. Grasp-cycle state machine. Labels (arm) and (hand) indicate which controller acts in each transition; the hand’s sensed-force registers gate the exit from CLOSE, and a failed VERIFY returns the cycle to PRE-GRASP for a bounded number of retries before the task is aborted.

to-pregrasp (arm), close-with-force-threshold (hand), verify contact (hand sensors), retreat (arm). The coordination primitive is a reliable completion signal for a hand command, provided by the actual-value registers. The latency of that signal over the TCI relay is a property the evaluation will measure (Section VI); it is not asserted here, because it depends on the relay and the serial configuration and has not been benchmarked.

The cycle is naturally expressed as the small state machine of Fig. 4, in which control alternates between the arm and the hand and the hand’s force registers gate the transition out of the closing state.

Fig. 5 gives the complementary view: where the state machine shows control states and the retry path, the sequence diagram shows the message exchange over time and, in particular, the points at which latency accrues. The register transactions marked t_{rt} are the ones tunnelled through the TCI relay, and their round-trip time is the quantity measured in Stage 1 (Section VI); the arm motions are commanded through a separate channel and are not affected by the relay.

E. Bill of Materials

Table VI lists the components the integration requires beyond the arm and the hand themselves. The list is short because the design deliberately reuses the robot’s own serial channel instead of adding a fieldbus interface. Quantities are per single-arm installation. No prices are given: they depend on supplier, machining route, and institutional procurement, and quoting figures that have not been obtained would not be meaningful.

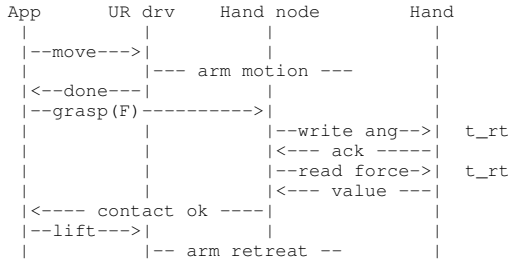


Fig. 5. Message sequence for a single grasp. Horizontal arrows are commands and responses; t_{rt} marks the register round-trips carried over the Tool Communication Interface relay, whose latency is measured in Stage 1. Arm motions use the driver’s own channel and are shown condensed.

TABLE VI
ADDITIONAL COMPONENTS REQUIRED FOR THE INTEGRATION

Item	Purpose / requirement
Adapter plate (1)	Mates EN ISO 9409-1-50-4-M6 to the hand wrist; machined or printed
M6 fasteners + locating pin	Flange-side mounting per EN ISO 9409-1
24 V supply, ≥ 5 A (1)	Concept B power source, sized above the 5 A peak
Strain-relieved power cable	Routed along the arm; conductor sized for 5 A peak
Twisted-pair RS-485 cable	Differential pair; shielded, with ground bond to tool connector
USB-RS-485 converter (1)	Fallback communication path, bypassing the TCI
Tool-connector mating plug	M8 8-pin, for the flange-integrated wiring
Control PC (Linux, 1)	Runs the ROS 2 driver, hand node, and <code>socket</code> relay

F. Safety Considerations

Adding a tool changes the safety case of a collaborative cell. Three measures are planned: updating the robot’s payload, TCP, and tool-shape settings so the e-Series force- and speed-limiting functions account for the hand; keeping the hand’s per-DOF force thresholds conservative when a person shares the workspace, which bounds finger forces independently of the arm’s safety system; and recording in the risk assessment that the hand holds its object after an emergency stop due to power-off self-locking, which avoids dropped loads but must be known to operators.

V. USE CASES

The combination of a sensorised five-finger hand and a precise collaborative arm enables application classes that a parallel gripper cannot reach. Five families are identified as realistic in the laboratory, ordered by increasing complexity. Each is stated as an enabled capability, grounded in Table III; none is claimed to have been performed yet.

Table VII makes the underlying contrast explicit. The comparison is kinematic and specification-based, not experimental: entries follow from the number of actuated degrees of freedom, the available opposition directions, and the presence of integrated sensing, and “two-finger parallel gripper” denotes the generic class rather than any particular product. The pattern is

TABLE VII
KINEMATIC CAPABILITY COMPARISON (ANALYTICAL, NOT MEASURED)

Capability	Parallel gripper	RH56F1
Actuated DOF	1	6
Opposition directions	One, fixed	Thumb vs. 4 fingers
Two-point pinch	Yes	Yes
Enveloping / wrap grasp	Two contacts	Multi-contact
Tripod (three-digit) grasp	No	Yes
Hold and actuate (trigger)	No	Yes
Communicative gesture	No	Yes
Distributed contact sensing	External	Integrated (8 pads)

that a parallel gripper is adequate wherever a task reduces to one opposition axis, and inadequate as soon as a task needs a third contact, differentiated finger roles, or contact information.

A. Grasp-Repertoire and Benchmark Studies

The most direct use is systematic grasping: reproducing grasp types from the taxonomy of Feix et al. [6]—power grasps, precision pinches, lateral (key) grasps, tripod grasps—on a set of everyday objects, and recording success rates and positioning tolerance per type. The taxonomy describes human grasps, and an underactuated six-DOF hand realises only a subset of them: types distinguished solely by independent finger abduction, or by fine in-hand repositioning, collapse onto a smaller set of reachable postures. The protocol therefore filters the 33 types down to those the hand can mechanically express, and reports coverage of that filtered set rather than of the taxonomy as a whole. Because the UR5e repeats poses to ± 0.03 mm [3], the arm can present the hand identically across trials, isolating the hand’s contribution from arm-positioning noise. The output is both a characterisation of the new hardware and a baseline for later projects.

B. Force-Controlled Grasping of Fragile Objects

The eight tactile sensors (30 N full scale) and configurable force thresholds [1], [2] allow grasps that regulate contact force rather than finger position: paper cups, fruit, or thin-walled consumables can be grasped by closing until a small threshold force is reached, independent of exact object size. A natural demonstrator contrasts position-controlled and force-thresholded grasping of deformable objects, with crush and slip as outcomes. This exercises exactly the sensing for which the tactile variant was ordered and reflects the argument that distributed touch underlies gentle manipulation [8].

C. Teleoperation and Demonstration Collection

Following DexPilot, AnyTeleop, and Open-TeleVision [7], [9], [10], the arm–hand system can be teleoperated from human hand tracking: a tracker estimates the operator’s wrist pose and finger configuration, the wrist pose is streamed to the UR5e as a Cartesian target, and the finger configuration is retargeted onto the six hand DOFs. Because Open-TeleVision already uses Inspire hands [7], published retargeting for this

hand family can be adapted rather than built from scratch. Teleoperation also turns the platform into a data-collection rig for imitation learning, a capability relevant to follow-up work even though policy training is out of scope here.

D. Human–Robot Interaction: Handover and Gesture

An anthropomorphic hand changes the robot’s social affordances. Two interactions follow. In object handover, the robot presents an item in a human-like grasp, detects the receiver’s tug as a drop in sensed finger force, and releases—a contact-based trigger a parallel gripper without tactile sensing cannot implement cleanly; the collaborative safety functions of the UR5e allow such handovers without fencing. In communicative gesture, pointing, waving, or a thumbs-up are expressible directly as posture vectors, useful in demonstrations and outreach.

E. Laboratory and Light Industrial Handling

Finally, the system can address handling tasks whose object geometry defeats two-finger grippers: cylindrical containers grasped with wrap grasps, irregular tools grasped at their handles, or trigger-operated devices such as spray bottles that require one finger to actuate a control while the remaining fingers hold the body. These reflect the manufacturer’s own positioning of the hand for laboratory automation [2]. Even a single such demonstrator—picking up a spray bottle, carrying it to a target, and operating its trigger with the index finger—would concretely exhibit a capability that none of the department’s existing grippers possesses.

VI. PROPOSED EXPERIMENTAL METHODOLOGY

This section specifies the evaluation to be carried out once the hardware is integrated. Each stage is tied to a claim made above, so that execution will confirm or refute that claim.

A. Stage 1: Interface Verification

Stage 1 validates the electrical and communication design of Section IV. The current draw of the hand is measured with a shunt or clamp meter during representative motions—free closing, force-thresholded grasping, and stall against a rigid object—to establish whether the 1.5 A budget of Concept A is realistic and which register settings respect it. In parallel, command round-trip latency and register-polling throughput are measured over (a) the TCI relay and (b) a direct USB–RS-485 connection, using identical Modbus transactions, to quantify the overhead of the flange-integrated path. Metrics: mean and worst-case current per motion class; median and 95th-percentile round-trip time; sustainable polling rate.

B. Stage 2: Grasp Benchmark

Stage 2 grounds the use-case analysis of Section V. A fixed protocol of grasp types from the taxonomy—at minimum cylindrical power grasp, spherical power grasp, precision pinch, lateral pinch, and tripod—is executed over a set of ten household and laboratory objects, with a fixed number of trials per applicable object–grasp pair. The arm presents the hand at a taught pregrasp pose; the hand closes with a fixed

TABLE VIII
PRINCIPAL INTEGRATION RISKS AND DESIGN MITIGATIONS

Risk	Cause	Mitigation
Tool over-current	Simultaneous high-force actuation exceeds 1.5 A (Concept A)	Register current limits; default to Concept B
Dropped serial frames	RS-485 noise along a moving arm	Twisted pair, shielding, checksum and retry
Common-mode fault	External-supply ground not bonded to tool ground	Bond supply 0 V to tool-connector ground
No serial traffic	Tool I/O left in analog-input mode	Enable communication-interface mode in PolyScope [4]
Object crush	Position control on a fragile object	Force-threshold grasp using tactile feedback [1]
Grasp held after E-stop	Power-off self-locking retains the object	Record in risk assessment; operator training

force threshold; success is defined as lifting the object and holding it for a set interval. Metrics: success rate per grasp type and object, dominant failure modes (slip, ejection, unstable contact), and tolerance to deliberate pregrasp offsets.

C. Stage 3: Tactile Demonstrator

Stage 3 evaluates Section V-B: force-thresholded versus position-controlled grasping of deformable objects (for example a paper cup, a compliant foam block, and an egg in a containment tray), with a fixed number of trials per condition and object. Metrics: crush rate, slip rate, and steady-state sensed contact force. Together the stages test both project questions—Stage 1 the interfacing question quantitatively, Stages 2 and 3 the use-case analysis empirically.

VII. INTEGRATION RISK ANALYSIS

Because the system has not been built, the risks below are identified from the design analysis rather than from incident logs; each is paired with a mitigation already reflected in the design. Table VIII summarises them. The dominant risk is electrical: under Concept A the tool-supply current limit can be exceeded by simultaneous high-force actuation, which is the primary reason Concept B is the default (Section IV-B). The remaining entries are standard for RS-485 field wiring and for collaborative tools, and none requires hardware beyond what the integration already specifies.

VIII. DISCUSSION

Two results of the design analysis are worth drawing out. First, the integration is well-supported at the interface level: the hand’s RS-485 side matches the e-Series Tool Communication Interface at the physical layer, and the vendor forwarding URCap and ROS2 driver provide a documented software path (Section IV-C) [4], [5]. This means the software effort concentrates on a thin register-level hand driver rather than on

transport plumbing. Second, the binding constraint is electrical, not informational: the 5 A peak against a 1.5 A tool supply is the single hard limit, and it is what motivates the external-supply default of Concept B. Framing the integration around this constraint is the paper’s main design contribution, and it generalises—any high-current end-effector attached through an e-Series tool connector faces the same budget.

A secondary observation concerns transferability. Because the constraint is a property of the tool connector rather than of this particular hand, the same analysis applies to any end-effector whose peak demand exceeds 36 W at the flange: the design space reduces to derating the tool by configuration or routing power externally. The variant-selection reasoning in Section III generalises similarly—where a vendor offers several bus options, the arm’s tool interface, not the application, is usually the deciding factor.

IX. LIMITATIONS

Several limitations are stated openly. The electrical figures are datasheet values; the true current profile of the hand under load must be measured before the flange-powered concept is used, which is why the external supply is the default. The hand has six coupled DOFs, far fewer than the human hand, so grasp types requiring independent finger abduction or fine in-hand re-orientation are outside its repertoire, and the use-case analysis is scoped accordingly. The RS-485 control path is polled and shared between commands and sensor reads, which bounds closed-loop reaction speed; the practical bound is left to Stage 1 rather than asserted. The ordered unit is a right hand only, so bimanual tasks are excluded. Finally, and most importantly, this is a design-stage paper: no system has been built and no measurement has been taken, so every quantitative statement about the integrated system remains to be verified, and the contribution is the integration design and the evaluation plan.

X. FUTURE WORK

The immediate next step is to execute the three-stage protocol of Section VI once the hardware arrives, beginning with the electrical measurement that gates the choice between the two power concepts. Beyond that, the platform supports two extensions that are natural but out of scope here: teleoperation-based demonstration collection using retargeting adapted from published Inspire-hand systems [7], and use of the collected demonstrations for imitation learning. Documenting the adapter geometry, the register-level driver, and the ROS 2 configuration as a reusable package for the department is also planned, so that subsequent projects on the same arm can reproduce the integration.

XI. CONCLUSION

This paper analysed, at the design level, how an Inspire Robots RH56F1-E2R-T1 five-finger dexterous hand can be attached to a Universal Robots UR5e, and what the combined system enables. Using vendor documentation, it characterised both platforms, showed that the hand’s RS-485 interface

matches the e-Series Tool Communication Interface and that the vendor URCap and ROS 2 driver provide the software path, and identified the hand’s 5 A peak draw against the tool connector’s 1.5 A supply as the binding constraint, for which an external supply or register-level current limiting are the two resolutions. It surveyed five use-case families grounded in the hand’s verified capabilities, specified a three-stage evaluation protocol, and analysed the principal integration risks. The contribution is an integration design and a pre-registered evaluation plan; executing that plan on the assembled hardware is the next phase of the work.

REFERENCES

- [1] Beijing Inspire-Robots Technology Co., Ltd., “The dexterous hand RH56 series user manual,” 2024. [Online]. Available: <https://en.inspire-robot.com/wp-content/uploads/2024/02/INSPIRE-ROBOTS-THE-DEXTE ROUS-HAND-RH56-SERIES-USER-MANUAL.pdf> (accessed Jul. 15, 2026).
- [2] Beijing Inspire-Robots Technology Co., Ltd., “RH56F1 series dexterous hand,” product page. [Online]. Available: <https://en.inspire-robots.com/product/rh56f1> (accessed Jul. 15, 2026).
- [3] Universal Robots A/S, “UR5e technical specifications,” data sheet, updated May 2025. [Online]. Available: https://www.universal-robot.com/manuals/EN/TechSheets/UR5e_techsheet_pdf_online/UR5e_techsheet_en.pdf (accessed Jul. 15, 2026).
- [4] Universal Robots, “Setting up the tool communication on an e-Series robot,” `ur_robot_driver` documentation. [Online]. Available: https://docs.ros.org/en/iron/p/ur_robot_driver/doc/setup_tool_communication.html (accessed Jul. 15, 2026).
- [5] Universal Robots, “Universal_Robots_ToolComm_Forwarder_URCap: utility to forward the UR tool communication interface (RS-485) to a remote PC,” GitHub repository. [Online]. Available: https://github.com/UniversalRobots/Universal_Robots_ToolComm_Forwarder_URCap (accessed Jul. 15, 2026).
- [6] T. Feix, J. Romero, H.-B. Schmiedmayer, A. M. Dollar, and D. Kragic, “The GRASP taxonomy of human grasp types,” *IEEE Trans. Human-Mach. Syst.*, vol. 46, no. 1, pp. 66–77, 2016, doi: 10.1109/THMS.2015.2470657.
- [7] X. Cheng, J. Li, S. Yang, G. Yang, and X. Wang, “Open-TeleVision: teleoperation with immersive active visual feedback,” in *Proc. Conf. Robot Learning (CoRL)*, 2024. arXiv:2407.01512.
- [8] R. S. Dahiya, G. Metta, M. Valle, and G. Sandini, “Tactile sensing—from humans to humanoids,” *IEEE Trans. Robot.*, vol. 26, no. 1, pp. 1–20, 2010, doi: 10.1109/TRO.2009.2033627.
- [9] A. Handa, K. Van Wyk, W. Yang, J. Liang, Y.-W. Chao, Q. Wan, S. Birchfield, N. Ratliff, and D. Fox, “DexPilot: vision-based teleoperation of dexterous robotic hand-arm system,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020, pp. 9164–9170, doi: 10.1109/ICRA40945.2020.9197124.
- [10] Y. Qin, W. Yang, B. Huang, K. Van Wyk, H. Su, X. Wang, Y.-W. Chao, and D. Fox, “AnyTeleop: a general vision-based dexterous robot arm-hand teleoperation system,” in *Proc. Robotics: Science and Systems (RSS)*, 2023. arXiv:2307.04577.
- [11] A. Billard and D. Kragic, “Trends and challenges in robot manipulation,” *Science*, vol. 364, no. 6446, art. eaat8414, 2019, doi: 10.1126/science.aat8414.
- [12] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, “Robot Operating System 2: design, architecture, and uses in the wild,” *Science Robotics*, vol. 7, no. 66, art. eabm6074, 2022, doi: 10.1126/scirobotics.abm6074.
- [13] D. Coleman, I. A. Şucan, S. Chitta, and N. Correll, “Reducing the barrier to entry of complex robotic software: a MoveIt! case study,” *J. Softw. Eng. Robot.*, vol. 5, no. 1, pp. 3–16, 2014, doi: 10.6092/IOSER.2014.05.01.p3.
- [14] International Electrotechnical Commission, “IEC 61158: Industrial communication networks — Fieldbus specifications,” IEC, Geneva, Switzerland. EtherCAT is standardised as Type 12.